Introduction to Networks

Task 1 : Draw a simple diagram (hand-drawn or digital) showing how your phone connects to Instagram, including your device, WiFi router, ISP, and Instagram server. Label each part and the connections.



Explanation

- When I open Instagram on my smartphone, my phone connects to the Wi-Fi router through a wireless connection.
- The router forwards my data to the Internet Service Provider (ISP), which provides internet access. The data then travels through the internet to Instagram's servers.
- The server processes my request and sends the required content, such as posts, photos, and videos, back to my smartphone.

Task 2 : List 5 devices you use or see daily that are part of a network (for example: smartphone, smart TV, WiFi router, laptop, smartwatch). For each, write what type of network it connects to (LAN, WAN, PAN, etc.).

Task 2: Devices and Their Network Types

A network connects different devices so they can communicate and share data. The following are five devices used in daily life and the types of networks they connect to.

No.	Device	Network Type	Explanation
1	Smartphone	LAN / WAN	A smartphone connects to a LAN through Wi-Fi or to a WAN through mobile data.
2	Laptop	LAN / WAN	A laptop connects to a local network through Wi-Fi or Ethernet and can access the internet through a WAN.
3	Wi-Fi Router	LAN / WAN	A router connects devices within a LAN and forwards traffic to external networks, including the internet.
4	Smart TV	LAN	A smart TV connects to the home LAN through Wi-Fi or Ethernet to stream online videos.
5	Smartwatch	PAN	A smartwatch commonly connects to a smartphone using Bluetooth, which forms a Personal Area Network (PAN).

- Different devices use different types of networks depending on their purpose.
- LAN connects devices within a local area, WAN connects networks over large geographical areas, and PAN connects personal devices over a short distance.

Task 3 : Explain the difference between the Internet and an intranet using an example from your college or a public WiFi you have used.

Task 3: Difference Between Internet and Intranet

1. Internet

The Internet is a global network that connects millions of computers and devices worldwide. It allows people to access websites, send emails, watch videos, and communicate with others. Anyone with an internet connection can access publicly available services.

Example: When I connect my phone to public Wi-Fi at a café and browse YouTube or Google, I am using the Internet.

2. Intranet

An Intranet is a private network used within an organization, such as a college or company. It allows authorized users to access internal information, files, notices, and services. Access is usually restricted to authorized members.

Example: If my college has a private student portal available only to students and staff for checking attendance, assignments, and internal notices, that portal is an example of an Intranet.

3. Key Differences

Basis	**Internet**	**Intranet**
Access	Publicly accessible services	Restricted to authorized users
Coverage	Worldwide	Usually limited to an organization
Purpose	Global communication and information sharing	Internal communication and resource sharing
Security	Depends on the service and security measures	Usually protected by access controls
Example	Google, YouTube	Private college portal

- The Internet is a global network used to communicate and access information worldwide, whereas an Intranet is a private network used within an organization.
- A college can use both the Internet for external access and an Intranet for internal academic activities.

Task 4 : Pick any one app you use (like WhatsApp, Zomato, or Flipkart) and describe, in 4-5 lines, how data travels from your device to their server and back when you perform a search inside the app.

Hint: Mention the devices and networks involved, and what happens when you hit 'search'.

Application Selected: YouTube

- When I search for a video on YouTube, my smartphone sends the search request through Wi-Fi to the router.
- The router forwards the request to my Internet Service Provider (ISP), which routes the data through the internet to YouTube's servers.
- The server processes the search and sends the matching video results back through the internet, router, and Wi-Fi to my smartphone.
- Finally, the YouTube app displays the search results on my screen.

Devices and Networks Involved:

- Smartphone — runs the YouTube application.
- Wi-Fi Router — connects the phone to the home network (LAN).
- ISP — provides internet connectivity.
- Internet (WAN) — carries data between my network and YouTube's servers.
- YouTube Server — processes the search request and returns the results.

What Happens When I Click Search?

1. I type a video name into the YouTube search bar.
2. The app sends the search request over the internet.
3. YouTube's server processes the request and finds relevant results.
4. The server sends the results back to my phone.
5. The app displays the videos on the screen.